VU AI HUB · LECTURE NOTES

Project Intelligent Systems

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Written by Inaaya and shared on Notion. Original page: cheddar-wall-f3b.notion.site/2df9801a9309803797cfd2afb73782ad

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Research methods - experiments

An algorithm is a recipe (program) to solve problems. Then can be evaluated in several ways:

1. Theoretical
   - In terms of the problem size (length of the input, depth of the tree, number of nodes in the graph etc.)
2. Experimentally
   - When it is too hard to analyze theoretically
   - Run it and measure

Experimental Analysis of Bots

Compared to algorithms you have a task with no perfect solution, because you don't know what problem you're solving.

You can measure how well bots play

- In games against other bots
  - Against a specific bot
  - Against another version of the same bot

You measure things like:

- Games won
- Points gained

Tips

Have a good research question

Good questions lead to useful, general, relevant and credible results

- It is not interesting to measure how fast the random bot plays, but it is interesting to measure how good the random bot plays

Choose your opponent

Real world opponent vs Bot

- Real human games
- Bot games
  - Human heuristic based
  - Learning based
- Random games

Investigate the performance of your bot

- Compare to human performance → real human games
- See how well it performs against other bots → play against many bots
- Do not just play against random to show your bot is best

FAIR principles

Your data should be:

Findable

- Upload your work on the desired platform

Accessible

- Upload your work in the desired format

Interoperable

- Balance your coding and experimentation time
  - Do not start with (over)optimizing the code
- Do not reinvent the wheel
  - Use existing implementations from libraries

Reusable

- Keep a record of your:
  - Code
  - Parameters, seeds, settings etc.
  - Generated datasets
- You should be able to rerun every experiment you did
  - Git?
- For the randomized algorithms report the mean or standard deviation (do not just pick the best results)

Ensure Comparability

Run speed benchmarks on the same machine

- without other tasks going on

Measure the same thing

- use a fixed evaluation procedure

Report the full story

- Also include failed experiments and anomalies (if something works but you don't know why)
- When reporting averages, report the number of experiments, report the standard deviation, significance
- When using a learning approach, use multiple models/trainings, do not just use the best one

Try to explain what you find

It is not enough to just show results: try to find patterns that give evidence/answer to you question

- Having no pattern is a pattern
- Make sure your results support your conclusions

Present your data in informative ways

Data presentation is a form of storytelling; each figure is a chapter

Use tables and pictures

- Use the one which suits your story the best in the paper, the rest can go in the appendix

Use good graphs

- Do not put too much information

Use caption for non-obvious aspects

Tie your paper to literature

Use it whenever possible with the best you can find and at references to your paper

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Your Project

Design

- Formulation of the research question
- Is there a solution at all?
- It needs to be concrete and narrow enough to finish in a month

Project Idea Assignment

- You design and implement an experiment
  - You can modify an existing bot (or implement your own)
  - You choose suitable baselines to compare with
    - usually bots to play against
  - You change some variable
  - You measure a dependent variable
    - often games won/lost
  - You analyse how the dependent variable changes
  - You perform statistical tests